SDC HACKATHON 2026 · DOMAIN 2 · ORBYTE SMART SME CHALLENGE

HOLDWISE

From a WhatsApp price enquiry to an approved quotation in under a minute, with a person in control.

A quotation module for the Orbyte SME hub · SDG 8: Decent Work and Economic Growth

Live: quote-lah-hardware-quotation-mvp.replit.app **github.com/aithubxmu/Holdwise**

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PRICE ENQUIRIES PILE UP ON WHATSAPP

> boss, nak 20 bag simen + 5 batang paip PVC 2 inci, hantar Bangi esok, berapa?

- The owner decodes Malay/English shorthand, looks up prices in a notebook or Excel, checks stock, adds delivery, types a reply.
- Minutes of skilled time per quote; replies come hours later or never.
- Wrong prices and out-of-stock items slip through.

1.1 million

MSMEs in Malaysia: 96.9% of all businesses, 69.7% of them micro firms [1]

7×

more likely to qualify a lead when a firm answers within an hour, yet only 37% do and 23% never answer [2]

~8 min

of owner time per quote today (our assumption, to be validated with shop owners)

[1] SME Corp. Malaysia, MSME Insights 2023/24 (DOSM data). [2] Oldroyd, McElheran & Elkington, "The Short Life of Online Sales Leads", Harvard Business Review, 2011.

2 · WHO IT IS FOR

MICRO AND SMALL SHOPS THAT SELL ON WHATSAPP

- **Ah Seng Hardware, Kajang** (our synthetic demo shop): owner, two staff, 40 products, 15 enquiries waiting.
- The same pattern in printers, caterers, aircon services and building suppliers: catalogue + price list + stock + delivery zones.
- Customers are walk-ins, regulars and contractors who expect a price fast.
- They already run their back office on **Orbyte**: people, work, eSign. Sales is the missing module.

Roles in the shop

ROLE	CAN DO
Owner	Approve big quotes, discounts, price changes; settings
Staff	Draft, fix and send routine quotes
Viewer	Read only

A FIVE-STEP PIPELINE, NOT A CHATBOT

1

TRIGGER

A WhatsApp enquiry lands in the Inbox (simulated).

2

DATA

Catalogue, prices, stock, customer type and delivery zones from the database.

3

DECISION

AI reads the messy text into items. **Rules price** them, apply discounts and delivery, check stock, flag doubts.

4

APPROVAL

Staff fix and send. Over RM 500, discount over 5% or a changed price: Owner only.

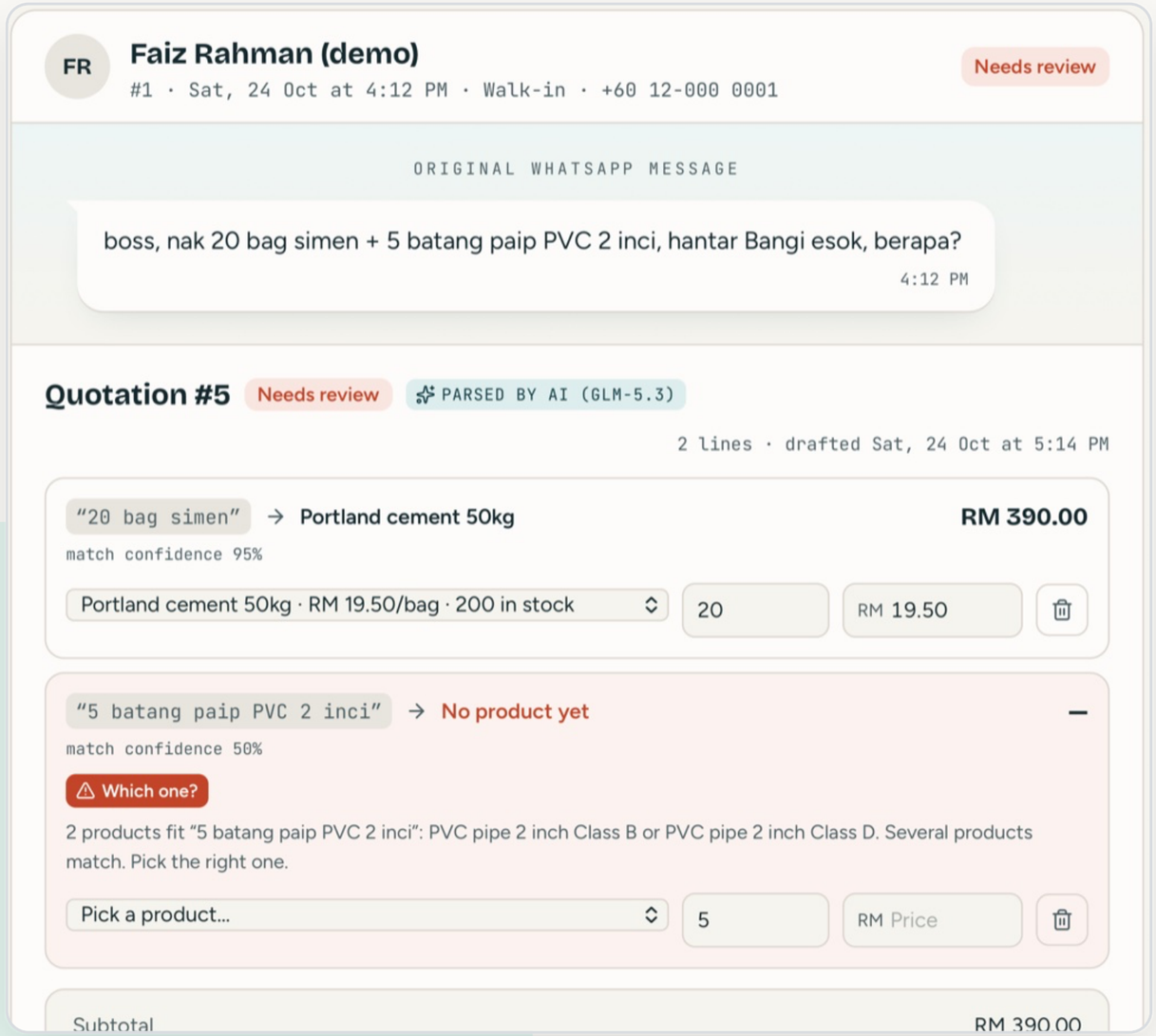
5

OUTCOME

Quote sent, stock reserved, 1 Orbyte credit charged, every step in the audit log.

Every quote is ready in under a minute, and nothing reaches the customer without a person's OK.

THE DRAFT: AI READ IT, A PERSON DECIDES



- **Parsed by AI (GLM-5.3)** badge on every draft; "rules (fallback)" with the reason when AI is off.
- "20 bag simen" → Portland cement 50kg, 95% sure, priced by rules: RM 390.00.
- "paip PVC 2 inci" fits Class B and Class D → **Which one?** The person picks.
- Then: Owner approval, simulated WhatsApp send, stock and credit updated.

5 · HOW IT WORKS

AI ONLY WHERE IT HELPS

AI: reading. GLM-5.3 on the sponsor's RunPod endpoint turns Malay / English / Manglish into strict JSON (items, qty, unit, confidence). Checked with Zod. **It never sees prices.**

Rules: money and stock. Catalogue matching, price × qty, discounts, delivery fee, stock check, approval routing.

People: decisions. Unknown item, two variants, confidence < 0.7, missing qty or low stock → a plain-language flag a person must clear.

No key, 8 s timeout or bad JSON → rules parser, and the draft says so.

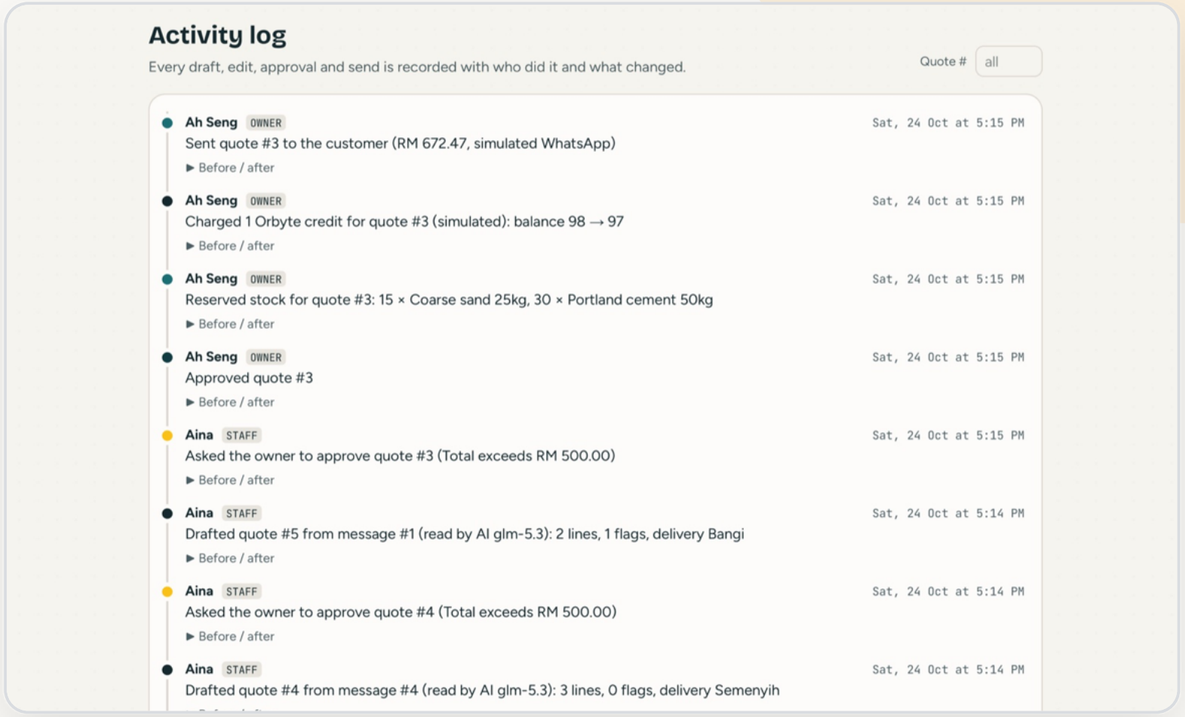
Measured on our synthetic test set (20 messages)

PARSER	ITEMS RIGHT	UNSEEN MSGS	TIME
GLM-5.3	37/37 (100%)	8/8	~4 s
Rules only	34/37 (91.9%)	5/8	<1 ms

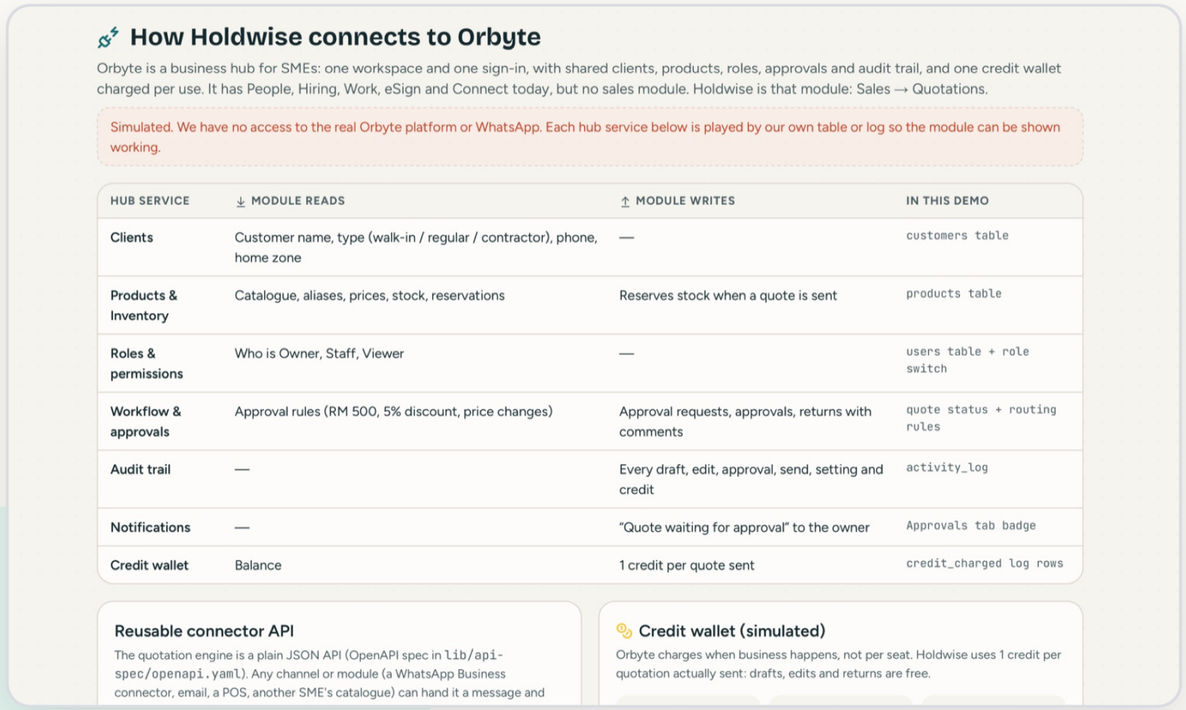
0 fallbacks. Small synthetic set: indicative, not a benchmark. Run it yourself on the AI eval page.

EVERY ACTION HAS A NAME ON IT

- Roles: Owner, Staff, Viewer. Viewers cannot change anything.
- Unresolved flags block sending; big quotes wait for the Owner.
- Activity log: who drafted, edited, approved, sent, with before/after.
- Synthetic data only; the AI gets message text and product names, never names, phones or prices.
- Phase 2 under PDPA 2010: consent notice, strip personal data before AI, fixed retention, role-based access.



A SALES MODULE THE HUB IS MISSING



How Holdwise connects to Orbyte

Orbyte is a business hub for SMEs: one workspace and one sign-in, with shared clients, products, roles, approvals and audit trail, and one credit wallet charged per use. It has People, Hiring, Work, eSign and Connect today, but no sales module. Holdwise is that module: Sales → Quotations.

Simulated. We have no access to the real Orbyte platform or WhatsApp. Each hub service below is played by our own table or log so the module can be shown working.

HUB SERVICE	↓ MODULE READS	↑ MODULE WRITES	IN THIS DEMO
Clients	Customer name, type (walk-in / regular / contractor), phone, home zone	—	`customers table`
Products & Inventory	Catalogue, aliases, prices, stock, reservations	Reserves stock when a quote is sent	`products table`
Roles & permissions	Who is Owner, Staff, Viewer	—	`users table + role switch`
Workflow & approvals	Approval rules (RM 500, 5% discount, price changes)	Approval requests, approvals, returns with comments	`quote status + routing rules`
Audit trail	—	Every draft, edit, approval, send, setting and credit	`activity_log`
Notifications	—	"Quote waiting for approval" to the owner	`Approvals tab badge`
Credit wallet	Balance	1 credit per quote sent	`credit_charged log rows`

Reusable connector API

The quotation engine is a plain JSON API (OpenAPI spec in `lib/api-spec/openapi.yaml`). Any channel or module (a WhatsApp Business connector, email, a POS, another SME's catalogue) can hand it a message and

Credit wallet (simulated)

Orbyte charges when business happens, not per seat. Holdwise uses 1 credit per quotation actually sent: drafts, edits and returns are free.

- Reads the hub's shared Clients, Products & Inventory and Roles; writes Approvals, Audit trail, Notifications.
- Credit wallet: 1 credit per quote actually sent; drafts and edits are free.
- Plain JSON API (OpenAPI spec): WhatsApp Business, e-mail or a POS could feed messages in.
- New industry = new catalogue + aliases; the pipeline stays the same.
- Integration is simulated in this prototype: no real Orbyte or WhatsApp access.

MINUTES INSTEAD OF HOURS, FEWER MISTAKES

−75%

staff time per quote
8 min → ≤ 2 min median, measured from the activity log

≥ 95%

item accuracy on unseen messages
held-out set of 50+ (today 8/8 GLM-5.3 vs 5/8 rules)

0

unapproved or mispriced quotes sent
100% of quotes approved by a person

SDG 8.2 Higher productivity through technological upgrading, in labour-intensive work: quoting is the most repetitive sales task of a micro retailer.

SDG 8.3 Support the growth of micro and small enterprises: faster answers win orders that are lost today.

Phase 2 targets from our Preliminary Proposal. Baseline of 8 min per quote is an assumption to be validated with shop owners.

WHAT COULD GO WRONG, AND WHAT IS NEXT

RISK	PLAN B
AI misreads a message	Catalogue-only matching, confidence flags, human approval
AI invents a price	Impossible: the model never sees or returns prices
Provider down or slow	8 s timeout → rules parser; switch provider by env var
Small test set	Phase 2: real, anonymised messages from partner shops

Phase 2 roadmap

WhatsApp Business connector · PDF quotes · 24 h follow-up drafts · second industry template · pilot with 2–3 shops

Business model

Sold through Orbyte: free tier, then pay per quote sent with Orbyte credits or a monthly plan.

THANK YOU

TRY IT NOW

Live prototype: **quote-lah-hardware-quotation-mvp.replit.app**
Code: **github.com/aithubxmu/Holdwise**

Built with Replit (Agent, Deployments, PostgreSQL, Secrets) and RunPod (GLM-5.3 Public Endpoint). Developed with help from Claude Code; declared in our Tech & Tools Declaration.

Team Holdwise: **Yerassyl Mubarak · Amina Shakhmetova**